Fiscal Year:	FY 2012	**Task Last Updated:**	FY 09/28/2012
PI Name:	Wise, John Pierce Ph.D.		
Project Title:	Mechanisms of Altered Gravity Potentiation of Metal-Induced Genotoxicity		
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Task Description:	The long-term objective of this research is to develop a model of the effects of microgravity on the cellular response to genotoxicity. The effects of microgravity on genotoxic response is important because NASA astronauts and personnel in microgravity will be exposed to significant genotoxic agents including chemicals (e.g. metals) and radiation and the potential for microgravity to exacerbate the genotoxic outcome is poorly understood. The short term objective of this research is to initially focus on chromium (Cr) because it is one of the only metals with a well-established DNA repair response to genotoxicity that is also a component of lunar soil. We will investigate the central hypothesis that altered gravity potentiates metal ion- and particle-induced genotoxicity. We will test this hypothesis through two Specific Aims: 1) Determine Altered Gravity Effects on Metal-induced DNA Damage Formation and Repair in Human Cells and 2) Determine if Altered Gravity Effects Are Conserved Across Cell Types and Species. These data when combined with our data generated by our studies of lunar dust in cells at normal gravity can be used to help determine appropriate exposure limits to moon dust. Our data will also be the first to evaluate how altered gravity affects metal-induced DNA repair gene expression and signal transduction as well as effects on metal-induced apoptosis. These data will also indicate which responses are general and which are cell or species specific. Altogether, these data should provide important insight into the mechanisms of altered gravity on metal-induced genotoxicity and suggest possible approaches for potential countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our study provides insight in how chromium(VI) , known human carcinogen, could affect human health on Earth as well as during space travel. This study defines a human cell-culture-based screening system that can be used to evaluate the risk of people who are exposed to altered gravity as the return to Earth. Our experiment with the bioreactor could provide information on developing a 3D culture system for further studies. Altogether, our data should provide important insight into the mechanisms of metal-induced genotoxicity in Earth and on space and suggest possible approaches for potential countermeasures. No innovative technologies developed.
Task Progress:	Summary of Research: To study the microgravity effect on cells, we developed a protocol for growing cells on microcarriers. We experimented different types of microcarrier beads, amount of microcarrier beads, cell type, seeding density of cells, types of environment to grow the cells on microcarriers (rocking or still), type of trypsin, trypsin time length, and type of rinse to produce the highest number of metaphases (metaphase index) during a chromosome harvest. These optimization experiments were done for both human lung and skin fibroblast cells (WTHBF-6, and BJhTERT, respectively). Once optimal growth and reagents were determined we were able to test growth of cells in simulated microgravity using the High Aspect Ratio Vessels (HARV) culture vessel bioreactor system. Based on previous studies, a rotation speed of 23 rpm was chosen to allow the cells to be in suspension while avoiding excessive centrifugal force. Quantify the effect of microgravity on Cr(VI)-induced genotoxicity Once all of these parameters were established we were able to begin genotoxicity studies. We found that cells grown on microcarriers and rotated at 23 rpm had less damage than cells grown at normal gravity under normal culture conditions. These data contradicted our previous data showing an increase in chromosome damage when microgravity was simulated by parabolas in a plane. We found that this difference was not due to cells growing on microcarrier beads. Because we saw more damage in flight, it occurred to us that we may not be properly modeling microgravity with a setting of 23 rpm. With the wide range of speed settings on the HARV culture vessel rotating base it is possible rotational speed may play a role in the chromosome damage results at 23 rpm in the previous experiments. To test for this we conducted experiments at 8 rpm. This speed is significantly slower than the 23 rpm speed used previously, but is still able to grow and maintain cells as demonstrated in previous studies (4-6). According to Hammond, 2001, as the rotation approaches zero the spirals of the HARV culture vessels are so large that the particles make one nearly continuous wall impact (6). At high rotation speeds, wall impacts increase again due to centrifugal forces (7). We found that sodium chromate induces even less damage at lower speed, further contradicting our flight data. We next considered that there could be an effect on Cr uptake. Our flight data showed less Cr uptake in microgravity leading to more damage. Thus, in order to further explain the possible differences seen in the amount of damage in cells treated in microgravity versus normal gravity, we conducted chemical uptake experiments to determine if the difference in damage was due to a difference in chromium uptake under differing gravity conditions. We found that cells in microgravity take up less chromium than cells under normal gravity conditions in human lung cells. These data were consistent with our flight data. When the chromosome damage data are considered based on Cr uptake we found that the amount of chromium-induced chromosome damage remains higher in normal gravity, in contrast to our published data of the effects of flight. We also conducted experiments in human skin cells and found similar results as those found in human lung cells. We conducted studies considering another Cr(VI) compound, lead chromate. Lead chromate is an insoluble form of chromate and is administered as a particle in suspension in cell cultures. There was no difference in the cytotoxic effect of lead chromate under microgravity and normal gravity conditions. We used the Western blot assay to consider specific proteins involved in DNA repair signaling. We focused on two major DNA double stand break repair proteins, Mre11 and Ku80. We chose these two proteins because DNA double strand breaks are one of the most hazardous DNA damage to the cell. Misrepaired DNA double strand breaks could lead to cancer formation. Two major repair pathways, non-homologous end joining (NHEJ) and homologous recombination (HR) are involved in the repair of this damage. Mre11 is a DNA damage sensor and facilitates HR. Ku 80 recognizes and binds broken DNA ends in NHEJ. Using western blot assays we found that Cr(VI) decreases Mre11 expression in cells under normal gravity and microgravity condition. The Mre 11 level in cells under microgravity condition is similar to the cells in normal gravity. Cr(VI) does not affect Ku80 protein expression in cells under normal gravity but slightly increased Ku80 in cells under microgravity condition. These data suggest that altered gravity does not significantly change cellular DNA double strand break response. Quantify the effect of hypergravity on Cr(VI)-induced genotoxicity In addition to working on the effects of microgravity we extended our investigation to consider the contribution of hypergravity using a centrifuge. We considered the effect of hypergravity on the cytotoxic effect of sodium chromate, a

soluble form of chromium, in human lung cells. Hypergravity exacerbated the effect of sodium chromate-induced cytotoxicity at lower concentrations in lung cells; however, at the highest concentration there was no difference in effect. We also looked at these effects in human skin cells. In skin cells there was no difference in the effect of hypergravity on sodium chromate-induced cytotoxicity. We also considered the contribution of hypergravity on the genotoxic effects of sodium chromate treatment. Our data show that sodium chromate induces similar chromosome damage at low concentration but less damage at higher concentration.

Determine if altered gravity effects are conserved across cell types and species

We investigated altered gravity effects in other species, specifically in bowhead whale and medaka cells. For these cells, we focused on the effect of hypergravity. There was significant time spent in working out the correct parameters and conditions of performing the experiments. Hypergravity was simulated using a centrifuge; however, we found that the centrifuge could not generate the temperatures required for proper culture conditions of the cells. The solution to this problem was to run the experiments at room temperature which involved additional testing to ensure that these variables were not influential in the final results. Growth curves were performed in which cells were exposed to room temperature for periods of time equal to those needed for the experiments. We found that this did not affect the overall growth of the cells and thus proceeded with some simple experiments to begin our investigations. We found that the cytotoxic effect of hexavalent chromium is not exacerbated by hypergravity in bowhead whale lung cells at 10g, however when we tested these cells at 2g we found a much greater difference.

We also considered the effects of chromate in medaka cells. We found that, similar to human skin cells, hypergravity did not exacerbate the chromium-induced cytotoxic effect in medaka cells. We then considered the genotoxic effect of chromium in medaka cells exposed to hypergravity. There was less chromium-induced damage seen in the cells exposed to hypergravity than cells exposed to normal gravity.

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